



# AgriShare

## Agricultural Equipment & Machinery Rental Marketplace End-to-End Visual Specification, Architecture Guide & Implementation Manual

**Executive Summary:** AgriShare is a high-performance agricultural machinery sharing platform engineered to eliminate capital expenditure barriers for smallholder and commercial farmers across India. It connects machinery owners with verified renters via a robust booking and escrow pricing engine.

### Core Architectural Pillars & Feature Modules

|                                                                                   |                                                                                              |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>TR</b> <b>Equipment Marketplace</b><br>Search & browse verified farm machinery | <b>AG</b> <b>Farmer / Renter Portal</b><br>On-demand reservations with zero upfront purchase |
| <b>CAL</b> <b>Booking Engine</b><br>Automated date locking & schedule management  | <b>INR</b> <b>Escrow Pricing Engine</b><br>Transparent daily rate & deposit calculation      |
| <b>REV</b> <b>Verified Reviews</b><br>Completed-booking gated reputation feedback | <b>NB</b> <b>Notification Hub</b><br>Real-time alerts for booking lifecycle transitions      |
| <b>DB</b> <b>Role Dashboards</b><br>Tailored views for Admin, Owner, and Renter   | <b>KEY</b> <b>JWT Security</b><br>Stateless authentication with token refresh                |

### Project Implementation Milestones

| Development Milestone         | Target Scope                    | Status    |
|-------------------------------|---------------------------------|-----------|
| Backend Architecture & Auth   | Full Stack Architecture & Tests | COMPLETED |
| Equipment Catalog & Search    | Full Stack Architecture & Tests | COMPLETED |
| Booking Engine & Pricing      | Full Stack Architecture & Tests | COMPLETED |
| Reviews & Notifications       | Full Stack Architecture & Tests | COMPLETED |
| Role-Based Dashboard          | Full Stack Architecture & Tests | COMPLETED |
| Modern Next.js Frontend       | Full Stack Architecture & Tests | COMPLETED |
| Indian Localization & Flat UI | Full Stack Architecture & Tests | COMPLETED |
| Automated Test Suite          | Full Stack Architecture & Tests | COMPLETED |

AgriShare ensures agricultural mechanization becomes universally accessible while providing equipment owners with steady recurring yields on idle machinery.

## Platform Overview & Design System

AgriShare tackles the critical disparity in farm mechanization by allowing tractor, harvester, and implement owners to monetize surplus machinery capacity while farmers acquire vital equipment on-demand without prohibitive capital loans.

### Platform Capabilities & User Experience

SEC

**Secure Authentication**  
Multi-role JWT auth & protected API routes

FL

**Faceted Filtering**  
Instant search by category, location, and price

CAT

**Equipment Catalog**  
Detailed machinery specs, deposits & availability

SCH

**Schedule Engine**  
Real-time availability calendar & conflicts lock

PRC

**Dynamic Pricing**  
Server-side daily rates + security deposit compute

REV

**Verified Reviews**  
1-to-5 star feedback verified via rental completion

NOT

**Event Notifications**  
Instant lifecycle alert triggers for all parties

DSH

**Analytics Dashboard**  
Revenue, fleet usage, and rental metrics

## Design System Specifications

| Design Token          | System Value / Specification           | Design Purpose                                     |
|-----------------------|----------------------------------------|----------------------------------------------------|
| Primary Color         | #236f41 — Forest Green                 | Agricultural identity, trust, and growth           |
| Accent Color          | #d97706 — Harvest Amber                | Action highlights, warnings, and alerts            |
| Surface & Light       | #f8fafc / #ffffff                      | Clean modern flat UI readability                   |
| Typography (Headings) | Times New Roman (Bold)                 | High-legibility editorial standard typography      |
| Typography (Body)     | Times New Roman (Regular)              | Clean classical readability across documentation   |
| Theme Modes           | Light / Dark Solid Theme Support       | Persistent theme preference stored in localStorage |
| Visual Language       | Solid Flat Cards with Subtle Hairlines | Crisp data presentation without heavy shadows      |

### Monorepo Architecture Layout

```
AgriShare/
├── backend/
│   ├── manage.py
│   ├── config/
│   └── apps/
│       ├── accounts/
│       ├── equipment/
│       ├── bookings/
│       ├── reviews/
│       ├── notifications/
│       └── dashboard/
├── frontend/
│   ├── app/
│   ├── components/
│   ├── services/
│   ├── lib/
│   └── scripts/
└── test_e2e_flow.py

# Django 5.1 + DRF Backend Application
# Management script
# Settings, ASGI/WSGI, root URLconf

# User roles, JWT auth, and profile management
# Machinery catalog, image storage, filters
# Booking lifecycle engine and conflict locks
# Verified post-rental rating and reviews
# In-app event-driven alert triggers
# Role-aggregated performance metrics

# Next.js 16+ App Router Frontend
# App routes: /explore, /dashboard, /bookings
# Flat UI components, Navbar, Hero, Machinery Cards
# Axios API clients with auto token refresh
# Utilities, formatters, and theme handlers

# Automated 10-step E2E integration test suite
```

## Machinery Fleet & Pricing Analytics

The data visualizations below represent the benchmark pricing and category distribution extracted from the registered Indian farm machinery fleet.

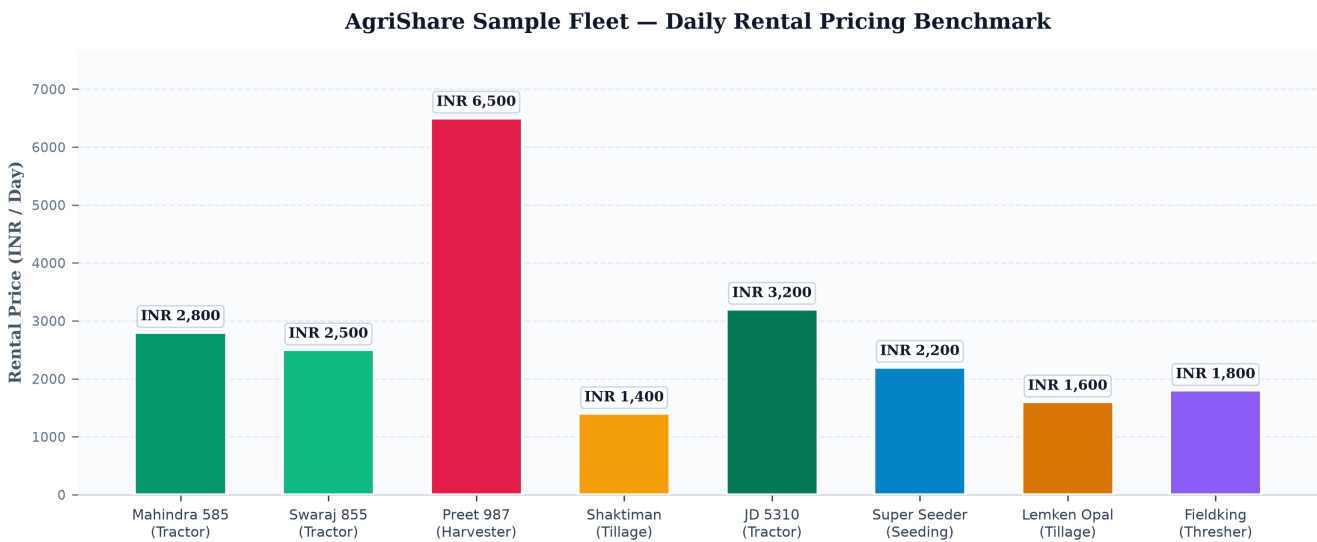


Figure 1: Daily Rental Rates Benchmark across Indian farm machinery listings (INR / Day).

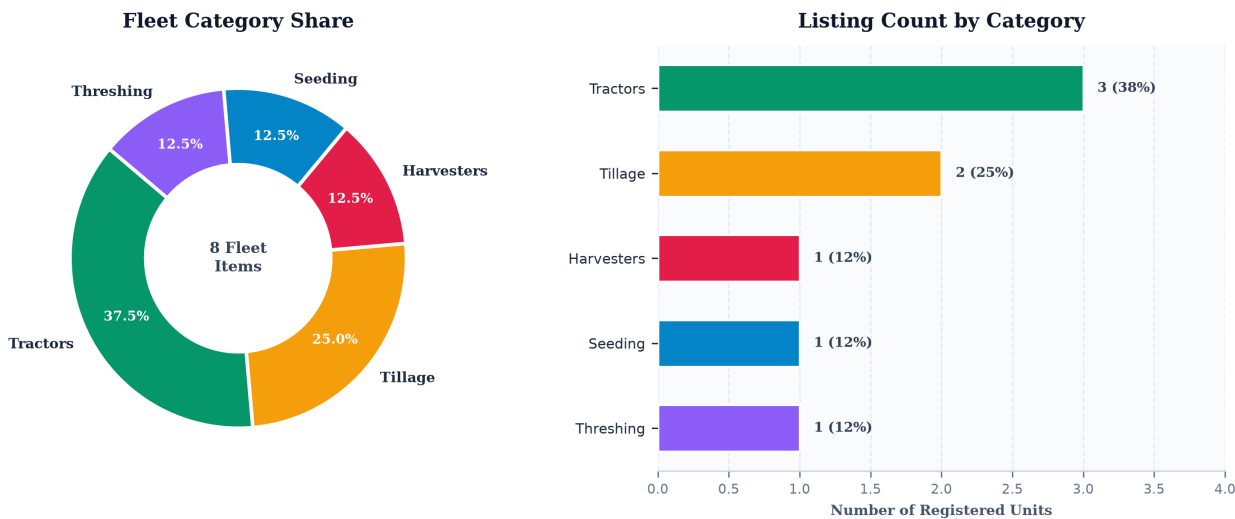


Figure 2: Machine Category Fleet Share (Donut) and Registered Units Count Breakdown.

## Booking Lifecycle & Rental Workflow

AgriShare enforces a strict server-validated state machine to govern machine rentals from initial reservation request to equipment return and escrow settlement.

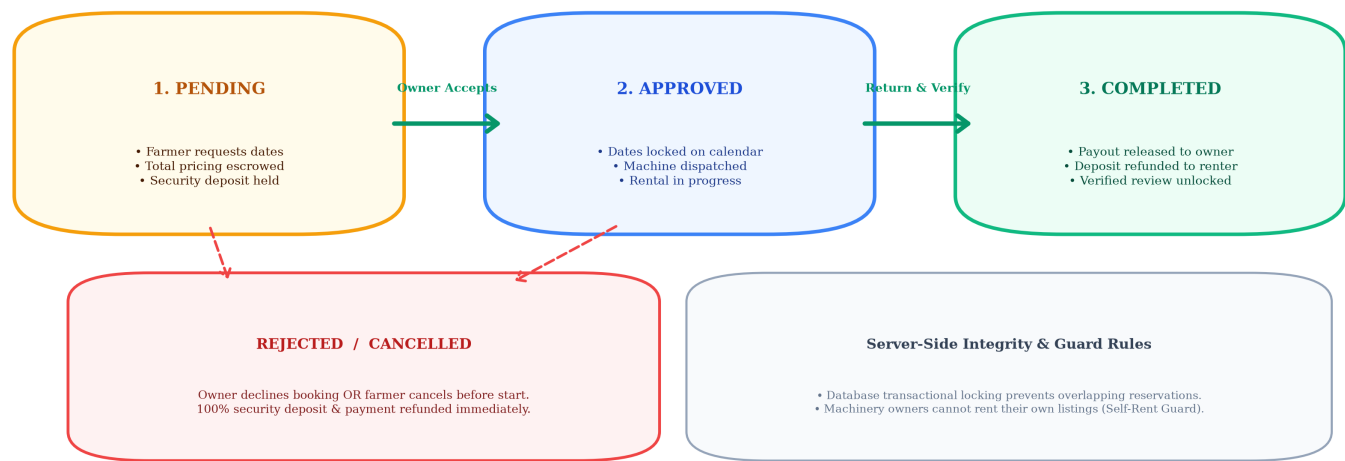


Figure 3: AgriShare State Machine: Primary approval pipeline and rejection/cancellation paths.

### Server-Side Escrow Pricing Engine

```
# Server-Side Total Pricing Calculation Formula:
Total Rental Days = (Rental End Date - Rental Start Date) + 1
Subtotal Amount  = Equipment Daily Rate (INR/day) × Total Rental Days
Total Escrow Due  = Subtotal Amount + Security Deposit (Refundable)
```

\* Security deposit is held in escrow and returned upon verified post-rental inspection.

### Verified Review & Rating Rules

- **Verified Rental Gating:** Reviews can ONLY be authored by renters whose bookings are in COMPLETED status.
- **Single Review Policy:** Exactly one review is permitted per booking ID to eliminate rating stuffing.
- **Dynamic Average Rating:** The machinery listing's aggregate rating and review count are recalculated synchronously.
- **Star Range:** Enforced integer rating between 1 (minimum) and 5 (maximum) with textual feedback.

### Notification Trigger Matrix

- **Owner Alerts:** Immediate alert on incoming booking request, cancellation before start, or review submission.
- **Renter Alerts:** Status updates upon owner approval, rejection, equipment dispatch, and escrow release.

## Technology Stack & Dedicated Brand Badges

AgriShare utilizes a decoupled modern web architecture combining a Python/Django backend API with a Next.js TypeScript frontend interface.

| Layer          | Technology Brand | Dedicated Logo Badge | Architectural Purpose                               |
|----------------|------------------|----------------------|-----------------------------------------------------|
| Frontend       | Next.js 16+      |                      | React App Router, SSR & Client Pages                |
| Language       | TypeScript 5+    |                      | Strict end-to-end type safety & interfaces          |
| UI Engine      | React 19         |                      | Declarative interactive state & reactive components |
| Styling        | Vanilla CSS      |                      | Clean solid Flat UI with light/dark theme tokens    |
| Icons          | Lucide React     |                      | Modern crisp vector icons across all UI modules     |
| Backend        | Django 5.1       |                      | Enterprise Python backend & ORM logic               |
| API Framework  | DRF 3.15+        |                      | RESTful endpoints with serialization & auth         |
| Authentication | SimpleJWT        |                      | Stateless JWT access & refresh token pair           |
| Query Filters  | django-filter    |                      | Declarative multi-param search & range filtering    |
| Documentation  | drf-spectacular  |                      | Automated OpenAPI 3.0 & Swagger UI schemas          |
| Media Engine   | Pillow           |                      | Server-side machinery image processing              |
| Primary DB     | PostgreSQL       |                      | Production ACID relational data storage             |
| Dev Database   | SQLite 3         |                      | Zero-config local development database              |

## Development Prerequisites

**Python 3.12+**  
Backend Django & DRF runtime

**Node.js 18+**  
Frontend Next.js compilation engine

**npm Package Manager**  
Client dependency management

**Git Version Control**  
Monorepo tracking & branch management

### Quick Startup Overview

```
Terminal 1 (Backend) → cd backend → source venv/bin/activate → python manage.py runserver
Terminal 2 (Frontend) → cd frontend → npm run dev
Browser Access      → http://localhost:3000 (UI) | http://127.0.0.1:8000/api/docs/ (API)
```

## Environment Setup & Demo Credentials

Follow these steps to initialize both server components for local development and demonstration.

### Backend Installation Commands:

```
cd backend
python -m venv venv && source venv/bin/activate # On Windows: .\venv\Scripts\Activate.ps1
pip install -r requirements.txt
python manage.py migrate
python manage.py seed_data # Seeds demo fleet & accounts
python manage.py runserver 127.0.0.1:8000
```

### Frontend Installation Commands:

```
cd frontend
npm install
npm run dev # Boots Next.js on http://localhost:3000
```

## Pre-Configured Demo Accounts

| Role Type | Role Logo Badge      | Email Address                  | Default Password | Operating Location     |
|-----------|----------------------|--------------------------------|------------------|------------------------|
| Admin     | <div>ADMAdmin</div>  | admin@agrishare.com            | admin12345       | New Delhi              |
| Owner     | <div>OWNOwner</div>  | gurpreet.singh@agrishare.com   | password123      | Ludhiana, Punjab       |
| Owner     | <div>OWNOwner</div>  | rajesh.patel@agrishare.com     | password123      | Rajkot, Gujarat        |
| Owner     | <div>OWNOwner</div>  | vikram.choudhary@agrishare.com | password123      | Karnal, Haryana        |
| Renter    | <div>RNTRenter</div> | ramesh.sharma@agrishare.com    | password123      | Indore, Madhya Pradesh |

## Registered Indian Farm Machinery Master Data

| Equipment Model               | Category  | Location               | Rate / Day (INR) | Security Deposit (INR) |
|-------------------------------|-----------|------------------------|------------------|------------------------|
| Mahindra Yuvo Tech+ 585 DI    | Tractor   | Ludhiana, Punjab       | INR 2,800        | INR 8,000              |
| Swaraj 855 FE                 | Tractor   | Sangrur, Punjab        | INR 2,500        | INR 7,500              |
| Preet 987 Combine Harvester   | Harvester | Karnal, Haryana        | INR 6,500        | INR 20,000             |
| Shaktiman Rotavator           | Tillage   | Indore, Madhya Pradesh | INR 1,400        | INR 4,000              |
| John Deere 5310               | Tractor   | Rajkot, Gujarat        | INR 3,200        | INR 9,000              |
| Dasmesh 912 Super Seeder      | Seeding   | Bathinda, Punjab       | INR 2,200        | INR 6,000              |
| Lemken Opal 090 Plough        | Tillage   | Nashik, Maharashtra    | INR 1,600        | INR 5,000              |
| Fieldking Multi-Crop Thresher | Threshing | Guntur, Andhra Pradesh | INR 1,800        | INR 5,000              |

## REST API Reference & Interactive Docs

AgriShare exposes a fully documented OpenAPI 3.0 specification accessible via interactive Swagger UI.

DOC

**Swagger UI Interactive Docs**  
<http://127.0.0.1:8000/api/docs/>

SCH

**OpenAPI JSON/YAML Schema**  
<http://127.0.0.1:8000/api/schema/>

### Core REST Endpoint Routes

```
# Authentication API
POST /api/auth/register/      # Register new User (Farmer / Owner / Admin)
POST /api/auth/login/         # Obtain JWT Access & Refresh token pair
POST /api/auth/token/refresh/ # Renew expired Access Token
GET  /api/auth/profile/       # Fetch authenticated user profile

# Equipment Marketplace API
GET  /api/equipment/categories/ # List all available machinery categories
GET  /api/equipment/            # Filtered list: ?category= &location= &min_price=
POST /api/equipment/           # Create machinery listing (Owner only)
GET  /api/equipment/{id}/       # Retrieve machinery specifications & rating
GET  /api/equipment/my_equipment/ # List caller's registered machinery fleet

# Bookings & Reservations API
POST /api/bookings/            # Create booking reservation with escrow
GET  /api/bookings/my_rentals/  # Fetch renter's active & past reservations
GET  /api/bookings/incoming/    # Fetch owner's incoming rental requests
POST /api/bookings/{id}/approve/ # Owner approves booking request
POST /api/bookings/{id}/reject/  # Owner rejects booking request
POST /api/bookings/{id}/complete/ # Complete rental & unlock review
```

## Automated Test Suite & Quality Verification

OK

**19 Django Unit Tests**  
Ran 19 tests in 1.4s — 100% OK

E2E

**10-Step E2E Integration**  
python scripts/test\_e2e\_flow.py — PASSED

NXT

**Frontend Production Build**  
npm run build — Zero TypeScript/Lint errors

SEC

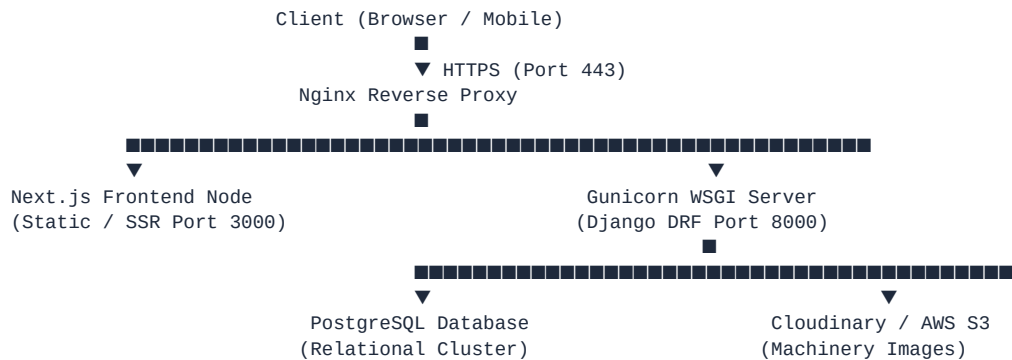
**Guard Integrity Verified**  
Overlapping bookings & self-rentals blocked

### Quality Assurance Checklist

- **ACID Transactions:** Booking approvals use select\_for\_update to eliminate race conditions.
- **Self-Rent Blocking:** Validation prevents machinery owners from creating reservations on their own assets.
- **Permission Scoping:** Role-based access controls strictly isolate Owner and Renter endpoints.

## Production Deployment Architecture

AgriShare is designed for cloud-native horizontal scalability with segregated application, caching, and storage layers.



### Production Security & Hardening Checklist

- **Environment Configuration:** Set `DJANGO_ENV=production` and `DEBUG=False`.
- **Database:** Migrate SQLite to managed PostgreSQL cluster with connection pooling.
- **Media Storage:** Connect AWS S3 or Cloudinary for reliable user machinery image storage.
- **CORS & CSRF:** Explicitly restrict `CORS_ALLOWED_ORIGINS` to verified frontend domains.

## Strategic Roadmap & Future Enhancements

|                                                                                                              |                                                                                                           |
|--------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <div>GPS</div> <div><b>GPS Live Discovery</b><br/>Real-time tractor telemetry &amp; field tracking</div>     | <div>MAP</div> <div><b>Map-Based Radius Search</b><br/>Geospatial radius queries around farmland</div>    |
| <div>PAY</div> <div><b>UPI &amp; Online Payments</b><br/>Instant payment settlement via Razorpay/UPI</div>   | <div>APP</div> <div><b>Native Mobile Apps</b><br/>React Native app for field offline access</div>         |
| <div>MSG</div> <div><b>Direct Chat Hub</b><br/>In-app messaging between owners and renters</div>             | <div>ID</div> <div><b>Aadhaar KYC Verification</b><br/>Government identity verification for safety</div>  |
| <div>AI</div> <div><b>AI Rental Recommendations</b><br/>Smart harvest scheduling based on weather data</div> | <div>IND</div> <div><b>12+ Indian Languages</b><br/>Localization in Hindi, Punjabi, Marathi, Telugu</div> |

## Licensing & Copyright

AgriShare is open source software released under the **MIT License**. Free to modify, distribute, and deploy for commercial and community agricultural applications.



# AgriShare

*Making Agricultural Machinery Universally Accessible, One Rental at a Time.*  
*Engineered with Django REST Framework • Next.js 16 • TypeScript • PostgreSQL*